The importance of DC link current in PMSM motor drives

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Hardware-in-the-loop (HIL) test systems are used to validate motor drive designs in the implementation phase and later. Permanent Magnet (Synchronous) Motor (PMSM) is the motor of choice in the automotive industry, for traction, power steering mostly. Engineers will want to validate the torque production and general behavior of their system using prototyped controllers first, then with production-level controllers, later, using HIL systems. This is generally called 'V-cycle' of system design.

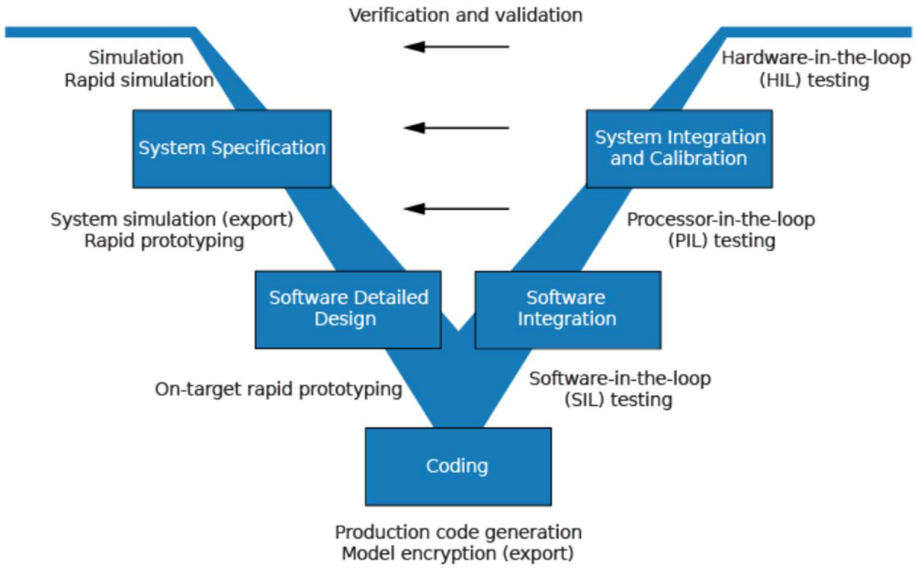

Fig. 1 V-cycle for system and controller development (Courtesy of The MathWorks)

Fault detection and system behavior under faulty conditions is also an important usage of HIL systems. HIL usage is even more important here because many real-world faults could damage a real motor drive, therefore it is really advantageous to use HIL systems for this. At the PMSM drive level, one very nasty fault is one where some IGBT is defunct. Malfunction could lead to an open-phase condition (mild fault) or a short-circuit, a severe fault because of counter-torque effects of the PMSM and possible destruction of the inverter. In the industry, it is common to verify IGBT fault by high-bandwidth monitoring of the DC link current and therefore, accurate simulation of this DC link current is critical.

This DC link fault detection principle is easy: every time a IGBT turns ON or OFF, there is a change of current on the DC-link, like a step. Therefore, for any IGBT gating action, if the corresponding current step is not detected on the DC-link, there is an IGBT fault. This most be done with less than 1 micro-seconds because of the fast current dynamic on these devices.

Because of the high bandwidth required, a latency of response in the order of 1 microsecond, FPGA-HIL systems are the best choice but they come with their own issues, in particular the difficulties of designing accurate electric system solver on FPGA. The Pejovic switch model, or constant admittance switch model, (a.k.a. Dynamic Switch model) is often used to design FPGA solver for motor drives. Again, the Pejovic switch model has its own problem if the alternating L/C equivalent state of the Pejovic switch is not properly taken care of.

Comparison of MathWorks Dynamic Switch and MATE Compensated Pejovic models.

We compare in this section the MathWorks FPGA-ready PMSM drive model of the page: https://www.mathworks.com/help/hdlcoder/ug/generate-hdl-code-for-simscape-models-using-dynamic-switch-approximation.html (as of April 5, 2026)

For convenience, this Mathworks and MATE models are accessible into the GitHub repo: https://github.com/KyleDiller/SimScape-Electrical-for-Real-Nice-Simulation

GitHub models:

sschdlexPMSMDynamicSwitchesREF.slx: Original Mathworks model with Dynamic switches
PMSMDynamicSwitches_init.m: Original initialization file of Mathworks
MATE_PejovicOverSamplingInverter.slx: : Mathworks model with MATE compensated Pejovic switches.

Result analysis

The comparison of the model can look somehow surprising to the inexperienced eye. While both models produce accurate motor currents, the DC-link current of the MathWorks model is completely wrong:

- 1: at the start of the simulation, before the motor starts, the DC-link current should be 0.
- 2: during the motor active drive phase, DC-link current should be in the order of the motor current themselves

The MATE model by comparison is totally accurate: null current before motor activation and DC-link current in line with the AC currents (common sense check). Validation with a detailed IGBT inverter (not shown) shows that both motor and DC-link currents are accurate in MATE.

Why is this Dynamic switch model so good for AC-motor and so bad for DC-link currents? This is related to the choice of the Pejovic admittance g=C/h=h/L (h is time step). Higher g means greater capacitance, lower g greater inductance. Motor is inductive but DC-link is capacitive. You cannot easily have it both ways. What is seen in the Mathworks case is the charging/discharging cycles of the Pejovic switch into the DC-link.

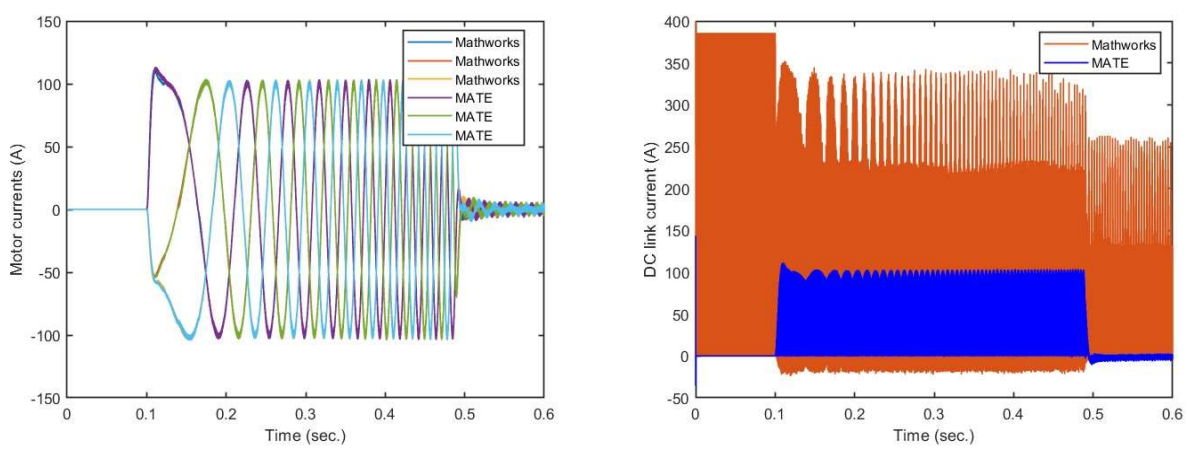

Fig 2. Comparison of MathWorks and MATE PMSM drive model: (left) motor currents, (right) DC-link current

Conclusion

Because the MATE Pejovic model has constant admittance, it is compatible with FPGAs and low-latency solver design. MATE accuracy is excellent. Examination of the model shows that proper initialization of the L/C state at transition of IGBTs level is very important to obtain this accuracy.

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